
Vision-Aware Autonomous Racing

A racing stack that laps on its own, reads its speed limits off trackside signs, dodges obstacles, and rescues itself when stuck.

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Four requirements, one catch

01

3 consecutive autonomous laps,
no teleoperation

02

Read ArUco speed signs — zones latch
until the next sign

SLOW 1.0

NORMAL 1.8

BOOST 2.5 m/s

03

Avoid obstacles with LiDAR; recover
from failures automatically

04

Evaluated on **hidden layouts** → nothing
can be hard-coded

Everyone publishes opinions. Only the arbiter drives.

```
v = min(v_curve, v_sign, v_lidar)
```

waypoint_manager

centerline waypoints + headings

path_follower

Pure Pursuit → steering

speed_planner

curvature → curve cap

sign_detector

ArUco → latched zone cap

lidar_safety

clearances → lidar cap + dodge bid

opinions

ROS topics

arbiter

takes the min,
blends authority,
runs the recovery
state machine,
publishes one drive
command



car

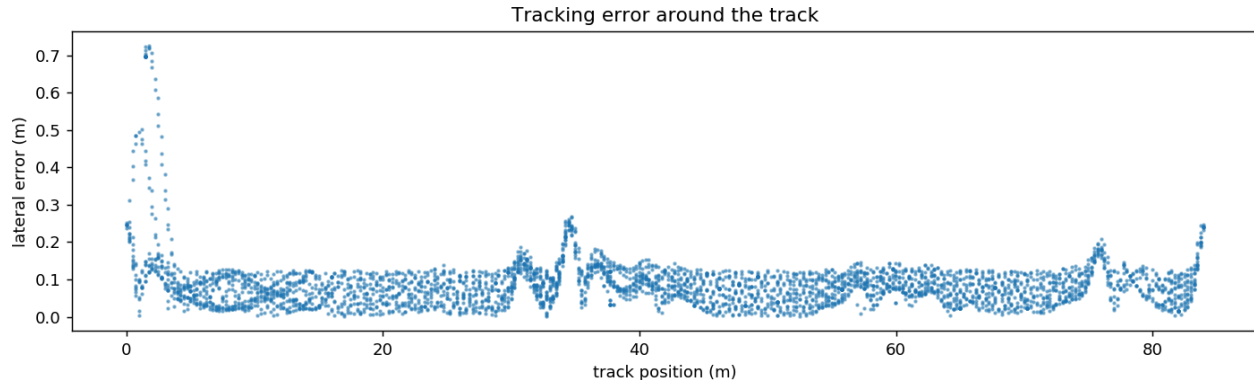
Geometry over regulation

Pure Pursuit chases a point 1.2 m ahead — one line of circle geometry gives steering.

No PID: the kinematics are known and the VESC layer beneath us already does the PID job.

Speed from curvature, taking the **minimum over the next 3 m** — so it brakes *before* the corner.

$$v = v_{\text{max}} / (1 + k|\kappa|)$$



finalDev_tracking_error.png — lateral error ≤ 0.24 m all lap (after the standing start)

Reading the rules off the wall

DICT_4X4_50

Binary-coded markers → near-zero false positives by construction.

OUR FILTERS

3 consecutive frames to confirm · largest marker wins · latch until a different sign confirms

PERTURBATION TEST

baseline	2 / 2 detected
-70 brightness	2 / 2 detected
7 px blur	2 / 2 detected
false transitions	0

The reflexes are a ramp, not a mode

720 → 3

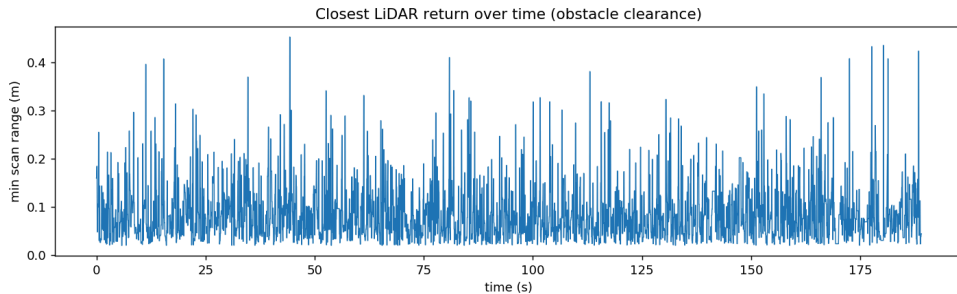
rays reduced to three clearances at the **10th percentile**, never the min — the laser hallucinates single-ray phantoms.

2.5 → 0.45 m

Linear speed cap: free at 2.5 m, **zero at 0.45 m**. E-stop is the ramp's bottom.

urgency

Dodge grows toward the freer side; the arbiter grants it full authority only near stopping distance.



finalDev_clearance.png — the raw minimum range is noise-dominated at ~0.1 m — exactly why the safety node trusts percentiles, never the min

Evaluation B put a box dead on the racing line

01 · symptom

Car stops safely... forever.

A safe deadlock: dodge was *designed* weaker than Pure Pursuit, and PP aims straight through the box.

02 · fix

Urgency-weighted authority blending in the arbiter.

Architectural, not a threshold tweak.

03 · bonus bug

Recovery was unreachable during a full e-stop.

The stuck detector required commanded speed > 0 — found while testing the fix.

04 · retest

B passes with zero recoveries needed.

Recovery proven separately: teleport-into-box → reverse → resume, on video.

08 — RESULTS

Pace is legality

Eval A spends 60% of the lap in SLOW zones — the car's speed is set by the signs it reads, not by tuning.

dev

46.5 s

0 collisions

eval A (unseen)

63.2 s

0 collisions

eval B (unseen)

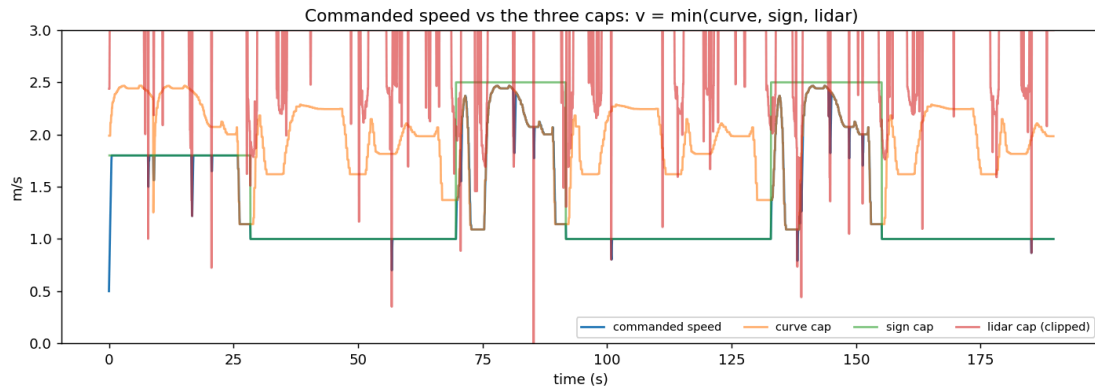
68.6 s

0 collisions

eval C (unseen)

80.9 s

0 collisions



finalA_speed_caps.png — the green sign cap sits at 1.0 m/s for most of the lap; commanded speed tracks it

09 — DEMO

Watch it drive

01

3 laps — counter ticking
to 3

02

Barrel dodge

03

Recovery: RACING →
REVERSING → RACING

Live option via RUNBOOK: cold start ~2 min · health check first · teleport-into-box command ready in a terminal.

Known edges

lap time

Speed-dependent lookahead + racing-line optimization — the biggest *legal* win left.

localization

Currently ground-truth odometry, switchable by parameter; drift is untested.

awareness

Safety windows are forward-only — no side or rear cones yet.

Questions.